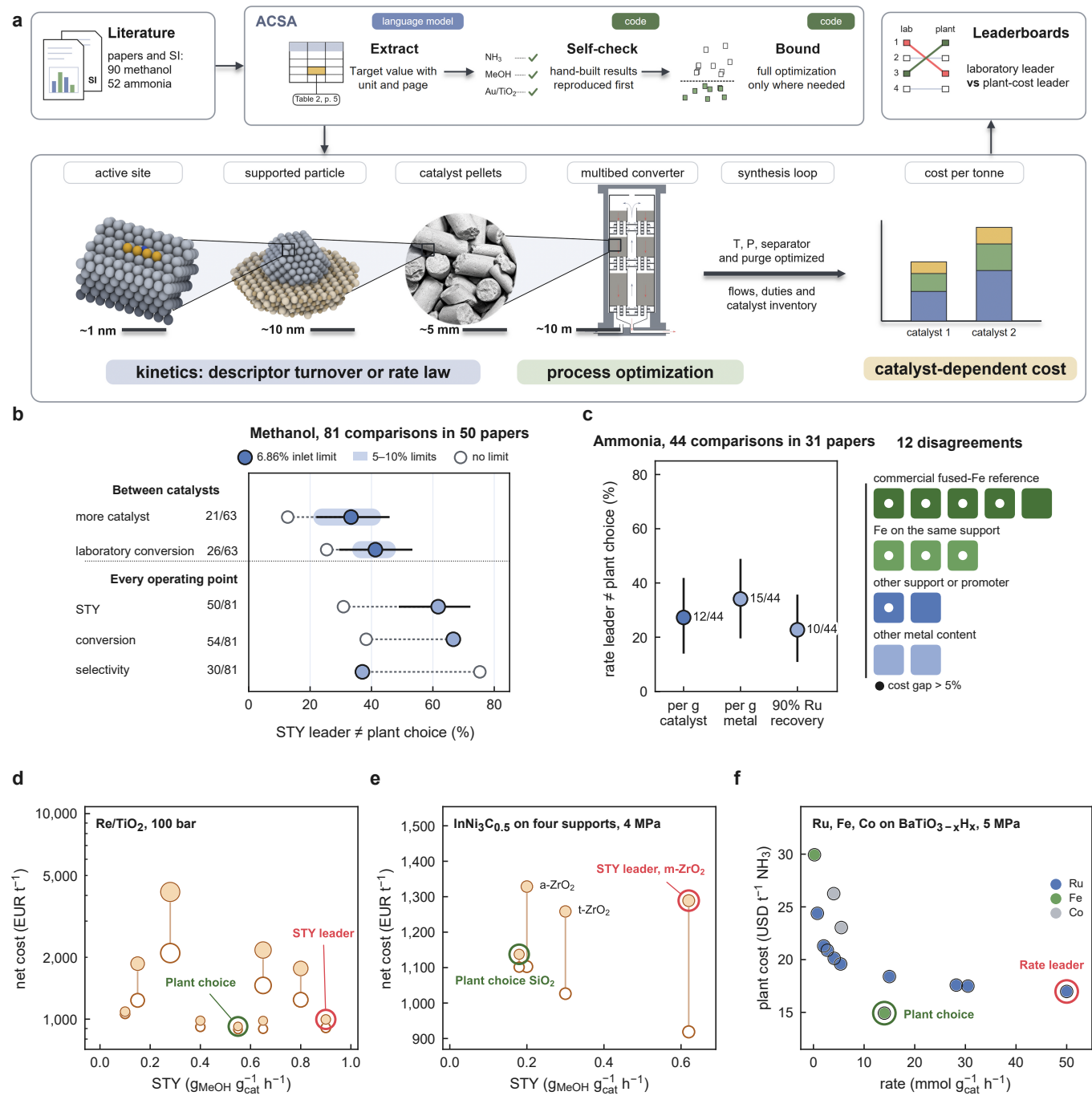

a
Literature
papers and SI:
90 methanol
52 ammonia
ACSA
language model
code
code
Extract
Target value with unit and page
Table 2, p. 5
NH3
MeOH
Au/TiO2
Self-check
hand-built results reproduced first
Bound
full optimization only where needed
lab
plant
Leaderboards
laboratory leader vs plant-cost leader
active site
supported particle
catalyst pellets
multibed converter
synthesis loop
cost per tonne
~1 nm
~10 nm
~5 mm
~10 m
T, P, separator and purge optimized
flows, duties and catalyst inventory
catalyst 1
catalyst 2
kinetics: descriptor turnover or rate law
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catalyst-dependent cost
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Methanol, 81 comparisons in 50 papers
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STY 50/81
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Ammonia, 44 comparisons in 31 papers
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per g catalyst
per g metal
90% Ru recovery
commercial fused-Fe reference
Fe on the same support
other support or promoter
other metal content
cost gap > 5%
d
Re/TiO2, 100 bar
net cost (EUR t−1)
STY leader
Plant choice
STY (gMeOH gcat−1 h−1)
e
InNi3C0.5 on four supports, 4 MPa
STY leader, m-ZrO2
a-ZrO2
t-ZrO2
Plant choice SiO2
net cost (EUR t−1)
STY (gMeOH gcat−1 h−1)
f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
plant cost (USD t−1 NH3)
Ru
Fe
Co
Rate leader
Plant choice
rate (mmol gcat−1 h−1)